
DETERMINANT

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1 Multiplicative Property

In this section we assume the usual definition of a determinant, from which one can derive the following properties that we will need:

$$\det \operatorname{diag}(d_1, d_2, \dots, d_n) = d_1 d_2 \dots d_n, \quad (1)$$

$$\det AB = \det A \det B, \quad (2)$$

$$\det \lambda A = \lambda^n \det A. \quad (3)$$

Now our task is to find all functions $f(A)$ from matrices to real numbers that satisfy the multiplicative property:

$$f(AB) = f(A)f(B). \quad (4)$$

We will use (4) repeatedly in the following calculation. For now we will make an assumption that the matrix A can be written as $A = PDP^{-1}$ where D is diagonal. Later we will try to generalize it for any matrix A using a Jordan form. We will also need matrices P_i that are unit matrices with the 1st and i -th rows permuted. Now we can compute:

$$f(A) = f(PDP^{-1}) = f(P)f(D)f(P^{-1}) = f(P)f(P^{-1})f(D) = \quad (5)$$

$$= f(PP^{-1}D) = f(D) = \quad (6)$$

$$= f(\operatorname{diag}(d_1, 1, 1, \dots))f(\operatorname{diag}(1, d_2, 1, \dots)) \dots f(\operatorname{diag}(1, \dots, 1, d_n)) = \quad (7)$$

$$= f(P_1 \operatorname{diag}(d_1, 1, 1, \dots) P_1^{-1}) f(P_2 \operatorname{diag}(d_2, 1, 1, \dots) P_2^{-1}) \dots \quad (8)$$

$$\dots f(P_n \operatorname{diag}(d_n, 1, 1, \dots) P_n^{-1}) = \quad (9)$$

$$= f(\operatorname{diag}(d_1, 1, 1, \dots) \operatorname{diag}(d_2, 1, 1, \dots) \dots \operatorname{diag}(d_n, 1, 1, \dots)) = \quad (10)$$

$$= f(\operatorname{diag}(d_1 d_2 \dots d_n, 1, 1, \dots)) = \quad (11)$$

$$= f(\operatorname{diag}(\det D, 1, 1, \dots)) = \quad (12)$$

$$= g(\det D) = g(\det PDP^{-1}) = g(\det A), \quad (13)$$

where we defined a function $g(x)$ using

$$g(x) = f(\operatorname{diag}(x, 1, 1, \dots)). \quad (14)$$

Let's compute the functional equation for $g(x)$:

$$g(xy) = f(\operatorname{diag}(xy, 1, 1, \dots)) = \quad (15)$$

$$= f(\operatorname{diag}(x, 1, 1, \dots) \operatorname{diag}(y, 1, 1, \dots)) = \quad (16)$$

$$= f(\operatorname{diag}(x, 1, 1, \dots)) f(\operatorname{diag}(y, 1, 1, \dots)) = \quad (17)$$

$$= g(x)g(y). \quad (18)$$

In the above computations we have only used (4) and no other assumptions.

Let's summarize our results. From (4) it follows that:

$$f(A) = g(\det A), \quad (19)$$

$$g(xy) = g(x)g(y), \quad (20)$$

for numbers x, y .

The equation (20) can be solved in several different ways.

Approach 1: With some extra smoothness assumptions one can show that the only solutions are $g(x) = 0$, $g(x) = x^s$ and $g(x) = |x|^s$. Consequently the only nonzero solution to (4) is a power of a determinant.

Approach 2: Instead of assuming some smoothness assumptions, we can instead assume homogeneity:

$$f(\lambda A) = \lambda^n f(A) \quad (21)$$

Using (19) and (20) we get:

$$f(\lambda A) = g(\det \lambda A) = g(\lambda^n \det A) = g(\lambda^n)g(\det A) = g(\lambda^n)f(A) \quad (22)$$

Comparing to (21) we get:

$$g(\lambda^n) = \lambda^n. \quad (23)$$

Since λ can be any real (or complex) number, we get:

$$g(x) = x \quad (24)$$

and from (19) we get:

$$f(A) = \det A. \quad (25)$$

Summary: assuming equations (4) and (21) we obtained (25) as the only nonzero solution. That means that we can define a determinant using (4) and (21) and no other assumptions.

TODO

The transition from λ^n to x in (23) might require better treatment, there might be some hidden assumptions about continuity on $g(x)$. Maybe there is a way to derive $g(x) = x$ more directly.

TODO

We assumed that the matrix A is diagonalizable. Remove this assumption by using Jordan form.

Roughly it will work as follows: if D is a Jordan form, then we get:

$$f(A) = f(PDP^{-1}) = f(P)f(D)f(P^{-1}) = f(P)f(P^{-1})f(D) = \quad (26)$$

$$= f(PP^{-1}D) = f(D) = \quad (27)$$

$$= f(\text{diag}(d_1 d_2 \dots d_n, 1, 1, \dots)) = \quad (28)$$

$$= f(\text{diag}(\det D, 1, 1, \dots)) = \quad (29)$$

$$= g(\det D) = g(\det PDP^{-1}) = g(\det A), \quad (30)$$

We can still use matrices P_i to bring all the values d_i to the (1,1) element, however the matrix $\text{diag}(d_1 d_2 \dots d_n, 1, 1, \dots)$ is not diagonal anymore, it will have some off-diagonal entries. The (1,1) element is still equal to $\det D$, because a determinant of a triangular matrix is just the product of elements on the diagonal. The matrix $\text{diag}(\det D, 1, 1, \dots)$ will have some off-diagonal entries. However, the determinant of such a matrix is equal to a diagonal matrix, so we can remove the off-diagonal entries (actually this is not clear if we can do it). One might be able to remove the off-diagonal entries by Gaussian elimination with similarity matrices. The rest of the calculation is the same.